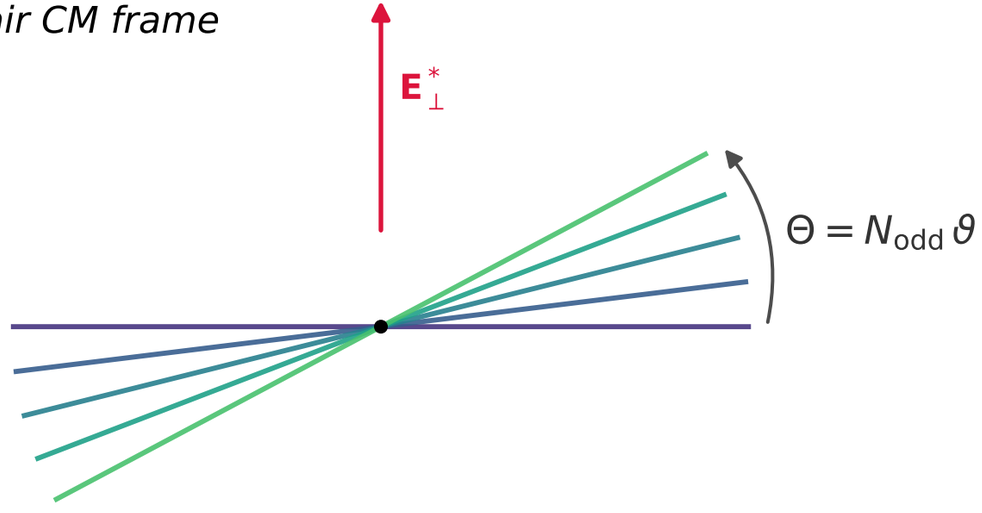
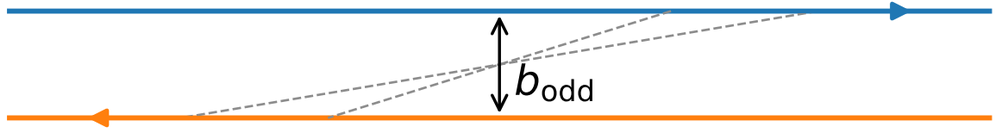
pair CM frame



axis after period 0, 1, 2, ...
(rotation exaggerated)

odd crossing: ends miss by $b_{\rm odd}$
even crossing: head-on ($b = 0$), no logarithm

transverse pull $\sigma b/|x|$



end 1 (+q)
end 2 (-q)

$b_{\rm odd} \ll |x| \ll \ell_{\rm max}$: $\int dx/|x| \simeq \ln(\ell_{\rm max}/b_{\rm odd}) + O(1)$